SPHOORTHY ENGINEERING COLLEGE

An Autonomous Institution Approved by AICTE & Accredited with NAAC Grade 'A'
Department of Examination & Academic Affairs · Hyderabad — 501510

Fluid Mechanics & Hydraulic Machinery: Bernoulli & Turbines

Program / Dept: B.Tech / MBA — MECHANICAL	**Course Code / Title:** Fluid Mechanics & Hydraulic Machinery (ME401PC)
Academic Regulation: Autonomous (SR23 / SR24)	**Curriculum Semester:** Semester 4 (Year 2)
Credits / Contact Hours: 3 Credits · 48 Lecture Hours	**Evaluation Scheme:** Continuous Internal (40M) + End Sem (60M)

Course Overview & Pedagogical Objectives:

Fluid statics (Manometry), Continuity, Momentum, Bernoulli's energy equation, Laminar & Turbulent pipe flow, Pelton wheel, Francis and Kaplan turbines. The primary objective of this course is to equip students with sound theoretical rigor, analytical problem-solving competencies, and real-world system engineering capabilities demanded by premier technological organizations and competitive graduate benchmarks.

Course Outcomes (COs) & Bloom's Taxonomy Mapping:

CO ID	Course Learning Outcome Description	Bloom's Level
CO 1	Understand and formulate fundamental concepts of Unit 1: Fluid Statics & Kinematics.	L2 Understand
CO 2	Analyze algorithms, mechanisms, and models in Unit 2: Fluid Dynamics.	L4 Analyze
CO 3	Design and evaluate robust architectures for Unit 3: Flow Through Pipes.	L5 Evaluate
CO 4	Implement and optimize operational pipelines in Unit 4: Hydraulic Impulse Turbines.	L3 Apply
CO 5	Integrate state-of-the-art frameworks in Unit 5: Reaction Turbines & Centrifugal Pumps for industrial solutions.	L6 Create

Comprehensive Module Index:

Unit	Title & Core Concepts	Page
Unit 1	**Unit 1: Fluid Statics & Kinematics**: Fluid properties (Viscosity, Surface tension, Capillarity), Pascal's law, Hydrostatic force on submerged plane...	Page 2
Unit 2	**Unit 2: Fluid Dynamics**: Euler's equation of motion along a streamline, Bernoulli's energy theorem derivation, Venturimeter, Orifice me...	Page 3
Unit 3	**Unit 3: Flow Through Pipes**: Reynolds experiment, Laminar flow between parallel plates (Couette flow), Hagen-Poiseuille pipe flow, Darcy-We...	Page 4
Unit 4	**Unit 4: Hydraulic Impulse Turbines**: Impact of fluid jets on flat and curved vanes, Pelton wheel component construction, Velocity triangles, Hydrau...	Page 5
Unit 5	**Unit 5: Reaction Turbines & Centrifugal Pumps**: Francis turbine, Kaplan axial flow turbine, Draft tube theory, Cavitation phenomenon, Centrifugal pump working...	Page 6

UNIT 1: FLUID STATICS & KINEMATICS • *Core Theory & Practice Guide*

1.1 Technical Exposition & Conceptual Foundations

The study of unit 1: fluid statics & kinematics forms an indispensable cornerstone of Fluid Mechanics & Hydraulic Machinery (ME401PC). Specifically, Fluid properties (Viscosity, Surface tension, Capillarity), Pascal's law, Hydrostatic force on submerged planes, Streamlines, Streaklines, Continuity equation. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

1.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 1 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

1.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 1
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

1.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 1 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 1. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 1 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 1 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 2: FLUID DYNAMICS • *Core Theory & Practice Guide*

2.1 Technical Exposition & Conceptual Foundations

The study of unit 2: fluid dynamics forms an indispensable cornerstone of Fluid Mechanics & Hydraulic Machinery (ME401PC). Specifically, Euler's equation of motion along a streamline, Bernoulli's energy theorem derivation, Venturimeter, Orifice meter, Pitot tube measurements, Momentum equation. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

2.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 2 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

2.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 2
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

2.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 2 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 2. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 2 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 2 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 3: FLOW THROUGH PIPES • *Core Theory & Practice Guide*

3.1 Technical Exposition & Conceptual Foundations

The study of unit 3: flow through pipes forms an indispensable cornerstone of Fluid Mechanics & Hydraulic Machinery (ME401PC). Specifically, Reynolds experiment, Laminar flow between parallel plates (Couette flow), Hagen-Poiseuille pipe flow, Darcy-Weisbach friction factor equation, Moody chart. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

3.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 3 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

3.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 3
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

3.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 3 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 3. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 3 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 3 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 4: HYDRAULIC IMPULSE TURBINES • *Core Theory & Practice Guide*

4.1 Technical Exposition & Conceptual Foundations

The study of unit 4: hydraulic impulse turbines forms an indispensable cornerstone of Fluid Mechanics & Hydraulic Machinery (ME401PC). Specifically, Impact of fluid jets on flat and curved vanes, Pelton wheel component construction, Velocity triangles, Hydraulic efficiency, Mechanical and overall efficiency. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

4.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 4 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

4.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 4
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

4.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 4 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 4. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 4 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 4 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 5: REACTION TURBINES & CENTRIFUGAL PUMPS • *Core Theory & Practice Guide*

5.1 Technical Exposition & Conceptual Foundations

The study of unit 5: reaction turbines & centrifugal pumps forms an indispensable cornerstone of Fluid Mechanics & Hydraulic Machinery (ME401PC). Specifically, Francis turbine, Kaplan axial flow turbine, Draft tube theory, Cavitation phenomenon, Centrifugal pump working, Manometric head, Specific speed of pumps. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

5.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 5 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

5.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 5
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

5.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 5 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 5. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 5 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 5 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

Standard Recommended Textbooks & Digital References:

1. *Primary Curriculum Standard Textbook* — Pearson Higher Education / McGraw-Hill Education (Latest Edition).
2. *Autonomous Reference Syllabus Blueprint* — Sphoorthy Engineering College Academic Council.
3. *NPTEL & IEEE Computer Society Guidelines* — Open Educational Resources & Peer-reviewed Publications.